

Data Warehousing

Overview & Architecture — from operational data
to strategic insight

What we'll cover

PART 1

Overview & Concepts

- Definition, working & types
- Characteristics & goals
- Components & evolution stages
- Users & how they query
- Real-world applications

PART 2

Architecture

- Architecture defined
- The design process
- Three-tier architecture
- Back-end tools & utilities
- ETL: extract, transform, cleanse, load

What is a data warehouse?

"A data warehouse is a **subject-oriented, integrated, time-variant**, and **nonvolatile** collection of data in support of management's decision-making process."

— W. H. Inmon

An environment, not a product. No single tool is "a warehouse" — it is a blend of databases, tools, and processes.

A read-mostly store. Data flows in, is cleaned and consolidated, then is queried — rarely changed once loaded.

How a data warehouse works

Operational systems capture transactions; the warehouse consolidates them into one historical, query-ready store.



MERIDIAN FOODS · Every night, POS logs from 400 stores load into the warehouse so buyers can compare this week's basket size against the same week last year.

Four defining characteristics

01 Subject-oriented

Organized around subjects — sales, customers, products — not around applications. Meridian models "sales," not "the POS app."

02 Integrated

Data from many systems is made consistent — one customer ID, one currency, one product code — across the whole enterprise.

03 Time-variant

Keeps years of history with a time key, so trends over months and seasons can be analyzed — not just today's snapshot.

04 Nonvolatile

Loaded data is read, not edited. New data is appended in bulk; existing records are not overwritten by users.



Goals of a data warehouse

A single version of truth

Consistent, reconciled data everyone can agree on.

Fast, easy access

Business users self-serve without writing complex joins.

Historical perspective

Analyze trends across years, not just the current state.

Separate analysis from operations

Heavy queries never slow down the checkout line.

Support decision-making

Turn raw transactions into strategic insight.

Consistent quality

Cleansed, validated data users can trust.

Types of data warehouse

EDW

Enterprise Data Warehouse

A centralized, enterprise-wide store spanning all subjects and business units. Meridian's company-wide sales, supply-chain and finance data in one place.

DATA MART

Data Mart

A subset scoped to one department or subject. A "Marketing mart" holds only loyalty and campaign data for that team.

ODS

Operational Data Store

Holds current, integrated data for day-to-day operational reporting, and often stages data on its way into the warehouse.

Data warehouse vs. data mart

DATA WAREHOUSE

Scope

Enterprise-wide, all subjects

Data

Deep history, fine-grained detail

Build approach

Top-down (Inmon)

Effort

Large, longer to deliver

DATA MART

Scope

Single department / subject

Data

Focused, often summarized

Build approach

Bottom-up (Kimball)

Effort

Smaller, faster to deliver



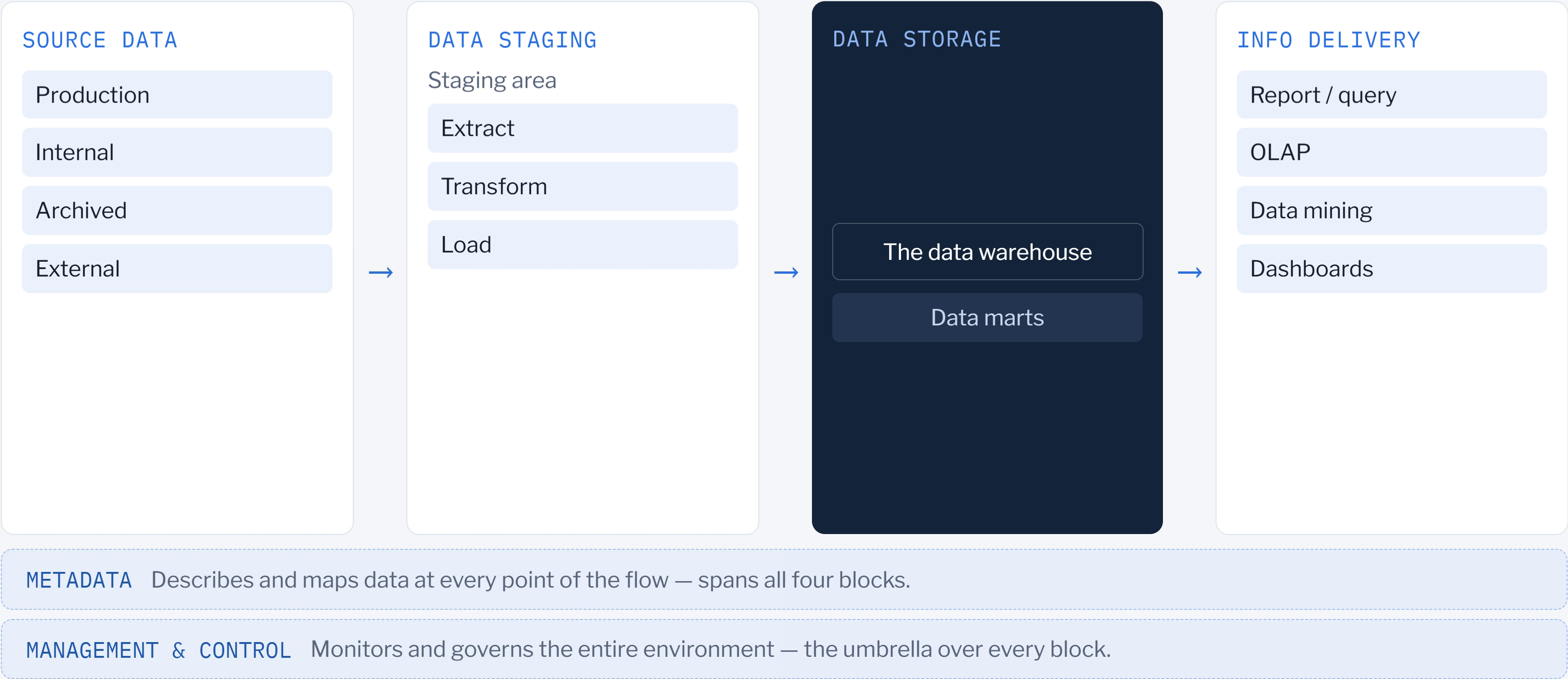
General stages of a data warehouse

Warehouses typically mature along this path as latency shrinks and integration deepens.

STAGE 1 Offline operational Reports run against a copy of the operational database.	STAGE 2 Offline warehouse Data loads on a regular schedule — nightly or weekly batches.	STAGE 3 Real-time warehouse Updated continuously as transactions occur in source systems.	STAGE 4 Integrated warehouse Feeds insight back into operations to trigger action automatically.
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Components of a data warehouse

After Ponniah, Fig. 2-7 — the building blocks and how data flows across them.



Who uses the data warehouse

EXECUTIVES Highly summarized KPIs & scorecards — the big picture at a glance.

MANAGERS Standard reports & dashboards to track their area against targets.

ANALYSTS OLAP slice-and-dice, ad-hoc queries & deep exploration.

POWER USERS Data scientists building models and mining the finest-grained detail.

How users query the warehouse

Reports & dashboards

Predefined, scheduled views of standard metrics.

"Weekly sales by region"

OLAP analysis

Slice, dice, drill down & roll up across dimensions.

"Drill from year → quarter → store"

Ad-hoc queries

Free-form questions users compose themselves.

"Which SKUs spiked in the heatwave?"

Data mining

Algorithms surface patterns humans wouldn't ask for.

"Diapers & beer sell together"

Applications across industries

RETAIL

Market-basket analysis, assortment & promotion planning.

BANKING

Fraud detection, credit risk & customer profitability.

TELECOM

Churn prediction & network capacity planning.

HEALTHCARE

Outcome tracking, cost analysis & resource planning.

MANUFACTURING

Supply-chain optimization & quality analysis.

GOVERNMENT

Program analysis, tax compliance & census reporting.



PART 2

Architecture

Architecture defined · the design process · the three-tier model · back-end tools & ETL

Data warehouse architecture

The **structural design** of the components and the **principles** governing the flow of data — grouped into three areas, wrapped by management and metadata (after Ponniah, Fig. 7-2).

MANAGEMENT & CONTROL — monitors & governs every operation

AREA 1

Data acquisition

Source systems → extract, transform, cleanse & stage.



AREA 2

Data storage

The warehouse database & marts, modeled for query speed.



AREA 3

Information delivery

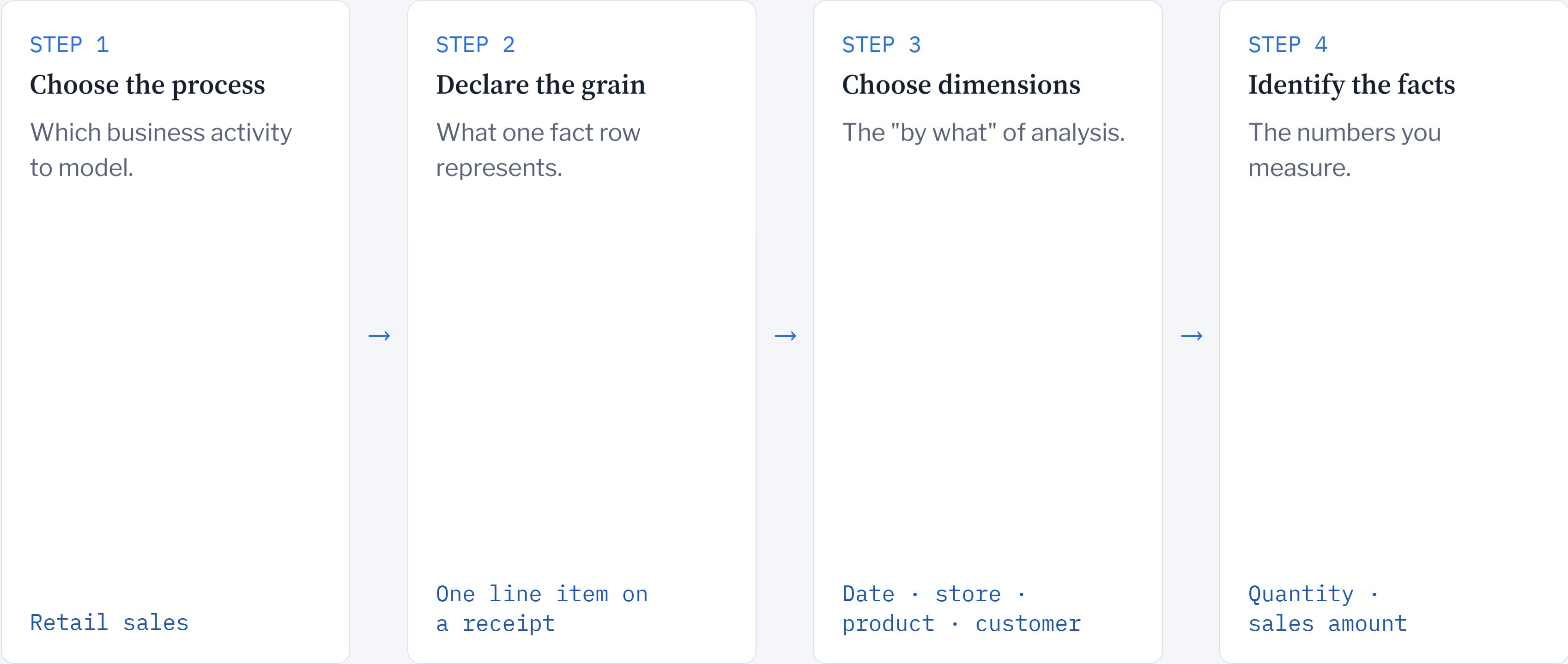
Reports, OLAP, dashboards & mining to end-users.

METADATA — the connective tissue that surrounds the entire flow of data



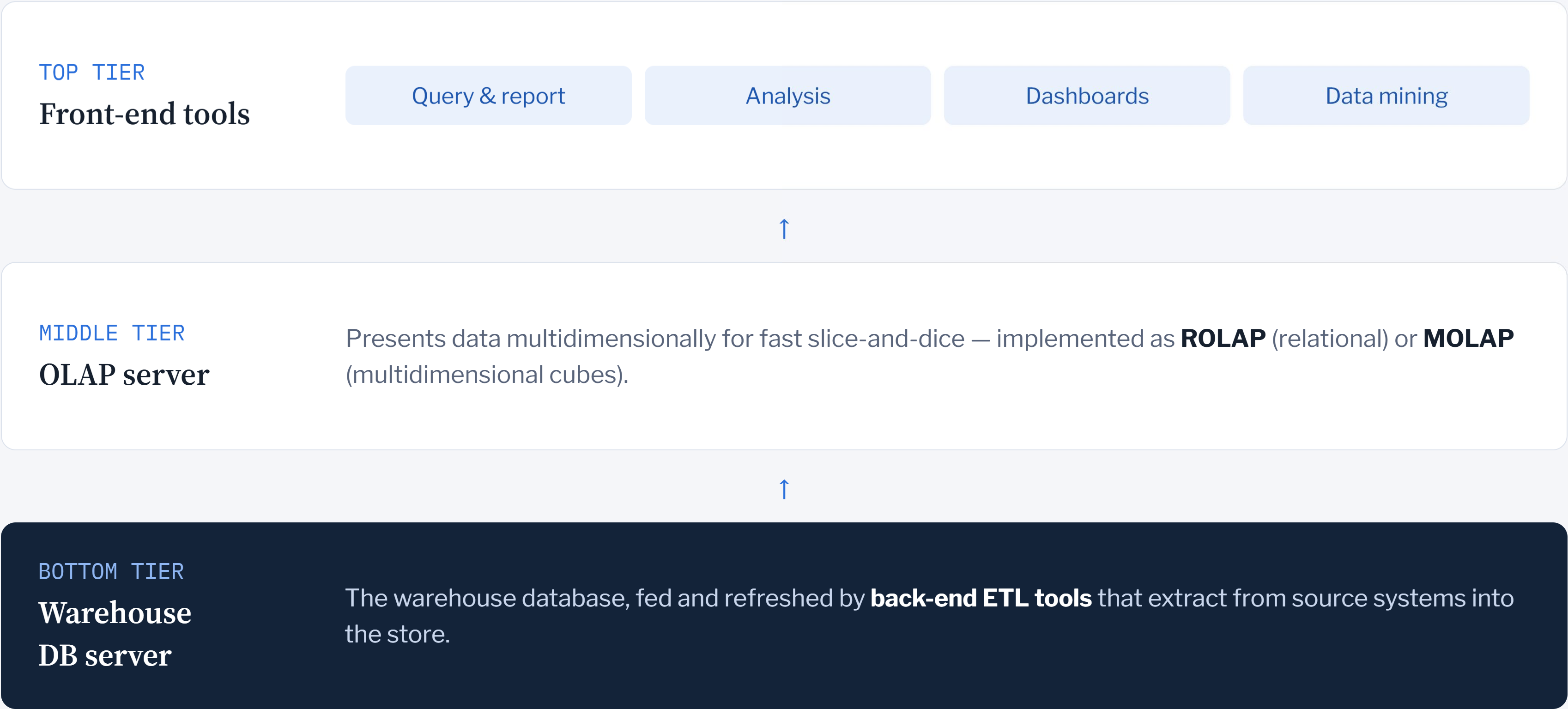
The design process

Dimensional design proceeds in four steps — illustrated with Meridian's retail sales.





Three-tier architecture



Back-end tools & utilities

The bottom-tier machinery that populates and maintains the warehouse — often the largest effort in the project.

EXTRACT

Pull data from heterogeneous source systems — flat files, databases, external feeds.

TRANSFORM

Convert, standardize, integrate & summarize data into the warehouse's common format.

CLEANSE

Detect & correct errors, de-duplicate records, and fill or flag missing values.

LOAD & REFRESH

Load into the warehouse, then refresh on a schedule with incremental updates.

ETL pipeline, step by step

Following one Meridian POS record from raw source to warehouse-ready fact.

1 · EXTRACT

Read the raw POS log from each store.

```
STR07,itm=0912,q=2,4.50,08/14
```

2 · TRANSFORM

Map codes to keys;
standardize date &
currency.

```
store=7 · sku=912  
2026-08-14  
· $9.00
```

3 · CLEANSE

Validate keys, drop
voids, resolve
duplicates.

```
sku 912 ✓ valid  
no duplicate ✓
```

4 · LOAD

Insert into the sales
fact table.

```
→ FACT_SALES  
row appended
```


Key takeaways

- 01 A data warehouse is **subject-oriented, integrated, time-variant & nonvolatile** — built for decisions, not operations.
- 02 Six components — source, staging, storage, delivery, **metadata & management** — move data end to end.
- 03 The **three-tier architecture** layers warehouse DB, OLAP server & front-end tools.
- 04 **ETL** — extract, transform, cleanse, load — is the back-end engine that keeps it all fed.